

Annotations Draft - G8906 Craft and Science

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Recipe 106r: Making gold run for casting

*Because gold cools down very quickly, you have to give it a mixture when it is well melted that will keep the heat. **Sublimate** softens it nicely. But, since it vanishes into smoke, it won't aid you for very long. Therefore mix this mixture into it when you want to cast it: of **sal ammoniac**, the best **verdigris** that you can find, a bit of **borax** and some **saltpeter**. And this, held over the heat, will become neat. The saltpeter makes it clear and heats it. But the key thing in this mixture is the verdigris, which has to be good. This mixture softens your gold so much that it becomes as tractable as lead, even if it is good gold.¹*

Because we had unique access to gold and a gold casting facility (and Dr. Boyd, an expert maker of sorts², to work with), we chose to work with two recipes relating to gold casting. We chose this recipe, along with 145r, because we felt they could be assayed easily (and fruitfully) in combination.

Recipe 106r, "Making gold run for casting," is a recipe for how to make a flux for gold casting and how to use it. There are three types of information in this recipe: the first is the information about the sublimate. The manuscript author tells us that sublimate (mercury sublimate, presumably), softens gold but dissipates too quickly to be truly useful. The second type of information is about the ingredients of the flux he recommends instead of sublimate: these are sal ammoniac (ammonium chloride), verdigris (copper acetate), borax (sodium borate), and saltpeter (potassium nitrate). Lastly, this recipe provides information about why and when to use the flux: it is necessary because gold "cools down very quickly," and it "softens your gold so much

¹ BnF Ms. Fr. 640, 106r.

² While it may seem unusual to refer to a dentist as an expert maker of cast gold objects, in fact Dr. Boyd has been using modern techniques to cast gold crowns, fillings, inlays and onlays for teeth for 50 years. Although today it is most common for dentists to outsource this type of laboratory work, Drs. Boyd, PC. is one of a handful of offices in New York City where the doctors do their own work in gold in order to insure quality.

that it becomes as tractable as lead”; and the flux should be used “when [the gold] is well melted.”

The author of the manuscript was very interested in recipes for what we now would consider fluxes, or recipes to make metals “run,” as he tends to describe it. The manuscript contains at least fifteen recipes that directly use the word “run” in reference to metals during casting, and there are others that mention various substances to add to metals while casting as well. Those that specifically speak of making metal “run” better prescribe adding various substances to copper, tin, brass, silver, and gold to improve their flow while casting, or simply describe the general necessity of using something to as a flux. The materials he suggests using as a flux for various metals include lead (015r, 047r), wallwort (016r), fixed quicksilver (069r), saltpeter mixed with common salt (101r, 116r), tin (111r, 116r), bismuth (115v), sal ammoniac (116r, 137r), leather (116r), mule’s feet (116r), calamine (116r, 136v), a mixture of orpiment, yellow arsenic copper filings, antimony, rock salt, tartar, glass gall, saltpeter, melted cooking salt, borax, and sal ammoniac (123r), a mixture of mercury sublimate and sal ammoniac (137r), a mixture of sal ammoniac, red copper and tin (154v). There are also two recipes, 106r and 120r, that refer specifically to a mixture of borax, sal ammoniac, salt peter and verdigris for use as a flux with gold. In fact recipe 120r, “Sandcasting for gold,” contains a section which is a direct parallel to 106r, in which he states “there is only” this mixture if your aim is to heat gold and keep it hot.³

³ The relevant section of recipe 120r reads: “Sublimate is often used by goldsmiths for gold. Some people add sulphur, but they are all wrong, because sulphur makes things sour, even if it heats [well]. And sublimate gets agitated, boils and bubbles. It is very good to clean gold because its exaltation draws [the bad stuff] out that disappears in the smoke. But to heat up gold, and to keep the heat, there is only the color of verdigris, salt of ammoniacum, saltpeter, and borax. This makes it runny and you will be able to throw in a branch from the Santoinge.”

It was difficult to research what other previous or contemporary makers used for flux, because there was not a specific word for “flux” at the time, only mentions of various materials being added to metals to improve casting practices. Hugh Plat seems fairly silent on the subject⁴, but Benvenuto Cellini mentions adding charcoal to lead, silver and copper when making niello to purify the mixture⁵. He also mentions using a combination of borax and verdigris as a flux for soldering silver.⁶ Borax has been used for nielloing in Europe since the 12th century, and is mentioned as a successful flux by several makers during the Early Modern period.⁷ Sal ammoniac and saltpeter were used as well during the Renaissance,⁸ but we were unable to find a reference to a recipe outside of BnF Ms. Fr. 640 that mentions using all four materials mixed together.

Questions that immediately arose for us upon first consulting this recipe included:

- What is “the best verdigris you can find”? What is “good verdigris”?
- What does it mean for gold (or for the flux) to “become neat”?
- What is “good gold”, and with what type of gold would our maker have been working?

⁴ Plat, Hugh, Sir. *The jewel house of art and nature: containing divers rare and profitable inventions, together with sundry new experiments in in [!] the art of husbandry*. London, 1653. *The Making Of The Modern World*. Web. 15 Dec. 2014.

⁵ Cellini, Benvenuto. *The Treatises of Benvenuto Cellini on Goldsmithing and Sculpture*. Trans. C.R. Ashbee. N.p.: Kessinger, n.d. Print.

⁶ Grieb, H. "Theophilus and the Shrine of Vitus: A Goldsmith's Technique in the Mirror of Ancient References." *National Museum of Australia Canberra ACT* (2004): n. pag. *National Museum Australia*. The National Museum of Australia. Web. 15 Dec. 2014.

⁷ Ibid.

⁸ Ibid.

We will suggest answers to these questions as we walk through our process for following the recipe.

The chief obstacle in creating this flux was acquiring the verdigris. All other ingredients were easily sourced from chemical companies. Verdigris proved possible to acquire from a company that sells traditional pigments for paints, but as we were unsure of how historically accurate the making of such a pigment might be, we decided to make our own.

We contacted Lisa Berry Drago, a doctoral candidate with the Chemical Heritage Foundation who is doing her research on the use of verdigris as a pigment in the 17th century. She suggested a basic recipe based closely upon a recipe found in the *Manuscripts of Jehan Le Bègue*,⁹ which essentially entailed creating a sealed environment in which un-oxidized copper plates were left to interact with distilled vinegar vapors for several weeks. A thin layer of vibrant green “efflorescence” appeared on the copper, which fell cleanly off when we simply bent the sheets. Three sheets of approximately 6x8 inches and one sheet of approximately 8x11 inches yielded about 8 ml of verdigris powder, which we then ground in a mortar and pestle to ensure there were no larger clumps. Ms. Drago suggested that perhaps for our maker the quality of the verdigris (i.e. “very good verdigris”) is be indicated by the color, and ours was certainly a very pure, very dark greenish blue. She also suggested that perhaps what the maker meant was a verdigris that was free from impurities, which we felt relatively confident that ours was. The third possibility she brought up was that the recipe might be referring to verdigris that was purified after being removed from the copper sheets by soaking it

⁹ "Verdigris." *Traveling Scriptorium*. N.p., 17 Jan. 2013. Web. 15 Dec. 2014.

again in vinegar, this we did not have time to do, given that our verdigris took so long to make as it was.¹⁰

To make the flux mixture, we combined ½ teaspoon of verdigris with ¼ teaspoon of borax, saltpeter and sal ammoniac, respectively. Because no specific amounts were specified by the recipe (except “a bit” of borax and “some” saltpeter), these calculations were left to our own discretion. We chose to add more verdigris than the other ingredients because the recipe specifically states that, “the key thing in this mixture is the verdigris.”

As far as the composition of the gold that we used was concerned, in the last sentence of the recipe, the maker writes, “This mixture softens your gold so much that it becomes as tractable as lead, even if it is good gold.” This seems to suggest that “good gold” indicates a more pure gold, which would be more difficult to melt because, not being alloyed with any metals that have a lower melting point, its melting point would remain quite high. Therefore, we felt safe that the flux specified in this recipe would work with whatever gold we had on hand. The gold we used was melted down from used crowns and inlays, and Dr. Boyd estimated that it was probably about 22 carat gold, although it was impossible for us to know for sure, as this was gold scrap that ended up in his office, not gold that his office had sourced for itself.

In practice, this flux was extremely effective. According to *Restorative Dental Practices*,

A flux is a substance applied to the surface of molten metal primarily to prevent oxygen in the air from coming in contact with the hot metal, to cause oxidation. In addition, the flux dissolves oxides, which may form while the metal is being heated...A flux in powder form is normally used for metals during the melting for a casting operation...The powder fluxes may contain finely divided charcoal or other ingredients mixed with borax and boric acid to give added protection by producing what is described as a reducing flux. The presence of the charcoal not only helps to

¹⁰ Berry Drago, Lisa. "RE: Lisa Berry Drago/CHF Verdigris Contact." Message to the author. 30 Oct. 2014. E-mail.

prevent the formation of oxides on the metal surface, but also tends to reduce to free metal any oxides that may have been formed. Fluxes in powder form may contain also a very small percentage (1 to 2%) of finely divided silica flour which aides in holding the molten flux in position on the surface of the hot metal.¹¹

A successful flux accomplishes two things: it must clear the surface of the molten gold of oxides, and additionally it must cause the gold to draw together into a rounder shape, indicating that the gold is in its purest form, and is ready to pour. The flux currently used in the dental office we were working in is composed of inorganic borates mixed with silica powder.

In our experiment, we used a propane torch to apply heat to the gold in a ceramic crucible. As the pieces of gold began melting, they drew together into a molten globule. However, this globule was somewhat flat and oxides were visible on its surface, giving it a soiled appearance. Once Dr. Boyd felt that the gold and the crucible were hot enough, we added our verdigris flux, which created an immediate change in the material: the flame around the crucible flashed greenish yellow, and a pungent, acidic odor arose from the crucible. The flux appeared momentarily to sizzle on the surface of the gold and the crucible, and then the oxide on the surface of the gold disappeared, and the gold did indeed draw together into more of a round shape.

The recipe itself states: “mix this mixture into it (i.e., the gold) when you want to cast it: of sal ammoniac, the best verdigris that you can find, a bit of borax and some saltpeter. And this, held over the heat, will become neat.” One of our questions when approaching this project was what the recipe means by “become neat.” At first glance, it

¹¹ Peyton, Floyd Avery. *Restorative Dental Materials*. F.A. Peyton with Associates, etc. (Second Edition.). C.V. Mosby Co.: Saint Louis, 1964. Print.

was unclear to us whether this referred to heating the flux on its own before applying it to the gold so that the flux “becomes neat”, or whether it meant that once the flux has been added to the gold and held in the heat altogether for a while, the gold will “become neat.” After attempting the process, we feel confident the phrase refers to the gold once the flux once has been added. We believe the phrase, “become neat” refers to the process (described above) of the gold drawing together into a rounder globule. Our physical experience of the materials indicates to us that the passage refers to the gold with the flux being heated together, not simply the flux alone.

In order to test how well the metal flowed, we decided to mold two different shapes that we thought might prove challenging for the gold to flow through, a small cross shape with bulbous ends, and a large unembellished ring. As a result of our early modern flux, we were able to pour our first mold, the small cross, perfectly on the first try. Our second mold, the ring, took three tries for the gold to flow completely through it, but eventually we were also able to get a successful cast of the shape.¹² This flux performed so similarly to the flux in use by the dental office that Dr. Boyd jokingly asked if it were available for purchase. According to him, “it worked exactly the same,” presumably aside from the flash of green and the acidic smell. It must be noted that while it very clear our verdigris flux worked, we are not sure precisely *why* it worked. It is possible that although our early modern maker emphasized the importance of the verdigris, the success of the flux was solely due to the borax, the ingredient common to both fluxes, and the other ingredients in the flux functioned simply as the silica flour does

¹² We believe that the two failed attempts at this casting were due to imperfections in our molds, and uneven application of heat to the gold and crucible, plus a pouring technique that improved with successive iterations, leading to a perfect cast on the third attempt.

in a modern flux, keeping the borax in place on the molten metal so it could do its de-oxidizing work.

We would like to conclude with recommendations for further inquiry about this recipe. One useful experiment would be to attempt to use each flux ingredient individually as a flux for gold, to ascertain which specific functions each chemical performs, or if there truly is something to be gained by mixing the components together in advance. This would be helpful to test the hypothesis that the only reason the early modern flux works well is because it shares the ingredient borax in common with the modern flux. The maker specifically states that the verdigris is the key ingredient in this flux. Further research must be done to ascertain whether he is correct in saying this, or whether perhaps this is due to some alchemical theory that has little bearing on the actual chemistry of the reaction that occurs with this flux when casting. Another experiment that would be useful would be to repeat our process exactly, but to use an early modern method of heating the gold, as it seems possible our propane torch (which achieves up to 3,623 °F) may have had something to do with the success we had with getting our gold to run, rather than the flux alone. A trial could be attempted where the flux is heated separately from the gold and prior to adding it to the gold. This would clarify definitively whether the flux was meant to “become neat” before being added to the gold or after. The flux could also be used with other metals, to gain a better understanding of why the early modern maker recommended this mixture as a flux specifically for gold, and for nothing else.